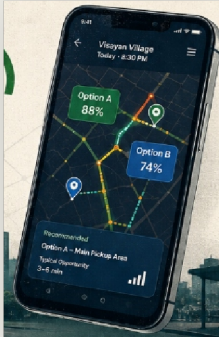


ROUTE SENSE

Know *where* rides are. Know *when* they're available.



- SMARTER DECISIONS**
Find the best places and times to ride.
- REAL-TIME INSIGHTS**
Up-to-date community mobility intelligence.
- CONNECTED JOURNEYS**
Know your ride. Know your mobile signal.
- PRIVACY FIRST**
Aggregated & anonymous on-device data.

1 THE PROBLEM

Knowing a route does not mean knowing if a ride is available right now.

Informal transit changes by time, traffic, and weather. Commuters face daily uncertainty:

- “Can I actually get a ride here right now?”
- “Where is the better place to wait?”
- “How reliable is transportation at this hour?”
- “Will I have mobile signal along the trip?”

2 BENEFICIARIES

PRIMARY USERS

- Students & daily commuters
- Workers & late-night commuters
- Delivery & gig-economy workers

SECONDARY BENEFICIARIES

- Schools, universities & transport coops
- Local government & transport planners
- Researchers & telco mobility planners

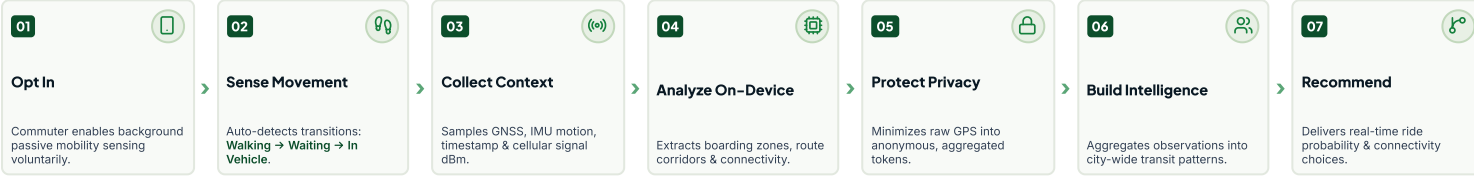
3 THE SOLUTION

RouteSense uses built-in smartphone sensors to understand informal transit behavior over time without driver apps or trackers.

- StopSense**
Learns pickup areas, movement, & service probability.
- SignalTrail**
Measures cellular connectivity along transit corridors.
- Where rides are available
- When they arrive
- How reliable estimate is
- Route cellular strength

4 HOW IT WORKS — PASSENGER SENSING TO MOBILITY INTELLIGENCE

7-Stage End-to-End Pipeline



5 KEY SYSTEM FEATURES

9 Core Capabilities

88% Service Probability
Live likelihood of finding a ride

Smart Pickup Areas
High-frequency boarding clusters

Opportunity Window
Realistic 3–6 min wait interval

Time-Based Trends
Morning, night & weekend patterns

Freshness Indicator
31+ observations • High

Connectivity-Aware
Compare speed vs cellular QoS

Pattern Discovery
Uncovers informal transit lines

Offline Caching
Full query access in dead zones

Zero Driver App
100% passenger crowdsensed

6 EMERGING TECHNOLOGIES & SENSORS

Zero vehicle installation • No driver app or GPS tracker required

Sensor Fusion & Activity AI
Fuses IMU + GNSS to detect Walking → Waiting → In Vehicle transit states.

Cellular QoS Mapping
Maps real-world carrier signal strength & connectivity dead zones along routes.

Spatial Clustering & ML
PostGIS + Uber H3 hexagonal binning with XGBoost predictive analytics.

Privacy-First Edge Inference
Processes movements on-device; publishes only aggregated, anonymous transit trends.

SMARTPHONE SENSORS UTILIZED

- GPS / GNSS
- Accelerometer
- Gyroscope
- Cellular Modem
- Magnetometer

8 REAL-WORLD RECOMMENDATION EXAMPLE

Visayan Village – 8:30 PM (Evening Peak)

Decision Engine: Compares arrival speed vs signal strength in real time.

CORRIDOR	AVAILABILITY	SIGNAL QoS	BEST FIT
National Highway	High (Every 3–5 min)	Moderate LTE	Fastest Commute

OPTION A: Main Terminal
Service Probability: **88%**
Typical Wait: **3–6 min**
Signal: **Moderate LTE**

OPTION B: Alternate Stop
Service Probability: **74%**
Typical Wait: **6–10 min**
Signal: **Very Good 5G**

FASTEST CHOICE: OPTION A (Save ~4 min)

BEST CONNECTED: OPTION B (HD Calls/Work)

7 TECH STACK

- Mobile Client: React Native + TS
- Sensor Layer: Kotlin Core
- Offline Cache: SQLite DB
- Mapping: MapLibre + OSM
- Backend DB: Supabase Postgres
- Spatial Index: PostGIS + Uber H3
- Data Science: Python + Pandas
- Inference: XGBoost ML

9 PROJECT TEAM & PROPONENTS

- Ryan Christian Deniega
- Kyndel Roy Suarez
- Dominice Aseberos
- Anjelo Vidal

CCE 106 — Application Development & Emerging Technologies | Student Researchers

10 REFERENCES & CITATIONS

- City Government of Tagum. City Ordinance No. 984, S. 2021 — Local Public Transport Route Plan.
- Android Developers. Activity Recognition Transition API & GnsMeasurement API.
- Android Developers. TelephonyCallback.SignalStrengthsListener API.
- Behrens et al. (2025). Spatial & temporal variability of informal transit using GPS observations.
- Travel Behaviour & Society (2025). Passenger smartphone GPS to identify travel, waiting, & transfer behavior.



LEARN MORE